

API Reference — AWS Invoice Pipeline

This document details the REST API endpoints provided by the **API Gateway** layer of the AWS Invoice Processing Pipeline. The endpoints route requests to the `approval-handler` Lambda.

Base URL

- **Deployed API Gateway URL:** `https://<api-id>.execute-api.<region>.amazonaws.com/prod`
- **Local Development (SAM local):** `http://localhost:3000`

Endpoints Summary

Method	Path	Description	Access
GET	<code>/approve</code>	Callback link from email to approve invoice	Public (signed links)
GET	<code>/reject</code>	Callback link from email to reject invoice	Public (signed links)
POST	<code>/approvals</code>	Perform approval, rejection, or reprocessing	Authenticated
GET	<code>/invoices</code>	List, search, and paginate processed invoices	Authenticated
GET	<code>/invoices/{invoiceId}</code>	Fetch details & audit logs for a single invoice	Authenticated
GET	<code>/exceptions</code>	List invoices requiring manual review / flags	Authenticated
GET	<code>/audit</code>	Query audit log history for a specific invoice	Authenticated
GET	<code>/dashboard</code>	Retrieve high-level operational metrics	Authenticated

Detailed Endpoints

1. Email Approval Callback (GET `/approve`)

Invoked when a reviewer clicks the **Approve** button in their review request email. Resumes the Step Functions workflow and redirects the user to an HTML success page.

- **Query Parameters:**
 - `token` (String, Required): Step Functions execution task token.
 - `invoiceId` (String, Required): The target invoice ID.
- **Response (200 OK):**
 - **Content-Type:** `text/html`
 - **Body:** Responsive HTML confirmation page.

- **Error Response (400 Bad Request):**
 - **Content-Type:** application/json
 - { "error": "Missing token or invoiceId" }
-

2. Email Rejection Callback (GET /reject)

Invoked when a reviewer clicks the **Reject** button in their review request email. Resumes the Step Functions workflow with a failure code and redirects the user to an HTML success page.

- **Query Parameters:**
 - **token** (String, Required): Step Functions execution task token.
 - **invoiceId** (String, Required): The target invoice ID.
 - **Response (200 OK):**
 - **Content-Type:** text/html
 - **Body:** Responsive HTML confirmation page.
-

3. Handle Invoice Action (POST /approvals)

Submits review decisions (Approve, Reject, Reprocess) from the React UI dashboard.

- **Request Headers:**
 - **Content-Type:** application/json
- **Request Body:**

```
{
  "invoiceId": "inv-10023-xyz",
  "action": "APPROVE",
  "taskToken": "AAAAKgAAAAIAAAAAAAAAAY...", // Optional: required only if resuming
  Step Functions
  "reviewer": "Reviewer Team A",
  "comments": "Totals matched. Standard approval given.",
  "correctedFields": {
    "gstIn": "27AAAAA1111A1Z1"
  }
}
```

- **Actions Supported:**
 - **APPROVE** : Marks status as **PROCESSED** and resumes workflow.
 - **REJECT** : Marks status as **RESOLVED** and halts workflow with rejection audit.
 - **REPROCESS** : Re-routes the invoice back into the Textract extraction pipeline.
- **Response (200 OK):**

```
{
  "success": true,
```

```
"invoiceId": "inv-10023-xyz",
"action": "APPROVE",
"message": "Invoice approved successfully"
}
```

4. Fetch Invoices (GET /invoices)

Lists all processed or in-progress invoices.

- **Query Parameters:**
 - `status` (String, Optional): Filter by status (`PROCESSED` , `IN_REVIEW` , `EXCEPTION` , `IN_PROGRESS` , `RESOLVED`).
 - `pageSize` (Number, Optional): Number of records to return (default: `25`).
 - `lastKey` (JSON String, Optional): Pagination token returned from the previous page's `lastKey` .
- **Response (200 OK):**

```
{
  "items": [
    {
      "invoiceId": "inv_101",
      "vendorName": "ACME Corp",
      "invoiceNumber": "INV-12345",
      "totalAmount": 1180.00,
      "status": "PROCESSED",
      "createdAt": "2026-06-24T12:00:00.000Z",
      "updatedAt": "2026-06-24T12:05:00.000Z"
    }
  ],
  "lastKey": "{\"invoiceId\":\"inv_101\",\"vendorId\":\"ACME Corp\"}"
}
```

5. Fetch Invoice Details (GET /invoices/{invoiceId})

Gets detailed metadata, extracted line items, anomalies, and audit log history for a single invoice.

- **Path Parameters:**
 - `invoiceId` (String, Required): Unique ID of the invoice.
- **Response (200 OK):**

```
{
  "invoice": {
    "invoiceId": "inv-10023",
    "vendorId": "PENDING",
    "vendorName": "Acme Software Solutions Ltd",
    "invoiceNumber": "INV-2026-001",
```

```

    "invoiceDate": "2026-06-20",
    "lineItems": [
      {
        "description": "Cloud Consulting Services",
        "quantity": 10,
        "unitPrice": 100.00,
        "amount": 1000.00,
        "hsnSac": "998311"
      }
    ],
    "subtotal": 1000.00,
    "cgst": 90.00,
    "sgst": 90.00,
    "totalAmount": 1180.00,
    "status": "PROCESSED",
    "extractionConfidence": 94,
    "s3ExtractedJsonKey": "textract/inv-10023/raw-response.json"
  },
  "auditLogs": [
    {
      "auditId": "a811c-9923...",
      "invoiceId": "inv-10023",
      "event": "AI validation completed",
      "eventType": "VALIDATION",
      "timestamp": "2026-06-24T12:02:00.000Z",
      "user": "system",
      "details": "Bedrock validation completed successfully."
    }
  ]
}

```

6. Get Dashboard Stats (GET /dashboard)

Retrieves statistics to populate KPI cards and trend charts on the landing page of the React UI.

- **Response (200 OK):**

```

{
  "total": 150,
  "processed": 120,
  "inProgress": 10,
  "exceptions": 15,
  "inReview": 5,
  "exceptionPercentage": "10.0",
  "monthlyTrend": [
    { "month": "June", "processed": 45, "exceptions": 3 }
  ],
  "vendorMetrics": [
    { "vendorName": "ACME Corp", "totalVolume": 15, "accuracyRate": "98.5" }
  ]
}

```

